

TOPIC No371224001
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' + \frac{1}{2x+1} y = \frac{2x-2}{\sqrt{y}}.$$

Problem 2. Find the general solution of the differential equation:

$$y'' - 4y' + 4y = (-3x - 2) \sin 2x.$$

Problem 3. Find the general solution of the differential equation:

$$\frac{y'}{\sqrt{x}} = 3y.$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y'' + 3y' = 9e^{-3x}, \quad y(0) = y'(0) = 0.$$

TOPIC No371224002
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Using the Laplace transform solve the Cauchy problem:

$$y'' + 2y' + y = e^{-2x}, \quad y(0) = y'(0) = 0.$$

Problem 2. Find the general solution of the differential equation:

$$y' = \frac{y}{x} - 2 \frac{x^2}{y^2}.$$

Problem 3. Find the general solution of the differential equation:

$$\frac{y'}{x^3} = y.$$

Problem 4. Find the general solution of the differential equation:

$$y'' - 4y = (x + 1) \sin 2x.$$

TOPIC No371224003
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Using the Laplace transform solve the Cauchy problem:

$$y''' + y'' = 0, \quad y(0) = 0, \quad y'(0) = 2, \quad y''(0) = -1.$$

Problem 2. Find the general solution of the differential equation:

$$y'' - 5y' + 6y = (2x - 2)e^{3x}.$$

Problem 3. Find the general solution of the differential equation:

$$y' - \frac{2}{x-1} y = (x-2)y^2.$$

Problem 4. Find the general solution of the differential equation:

$$y' = \frac{y^3}{x}.$$

TOPIC No371224004
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' - 2y = \frac{x+1}{\sqrt{y}} e^{3x}.$$

Problem 2. Find the general solution of the differential equation:

$$y' - 2y^3 = 0.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y''' - 8y'' + 16y' = 0, \quad y(0) = -3, \quad y'(0) = -4, \quad y''(0) = 16.$$

Problem 4. Find the general solution of the differential equation:

$$y'' - y = (2x - 2)e^{-x}.$$

TOPIC No371224005
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Using the Laplace transform solve the Cauchy problem:

$$y''' + 6y'' + 9y' = 0, \quad y(0) = 2, \quad y'(0) = 6, \quad y''(0) = 9.$$

Problem 2. Find the general solution of the differential equation:

$$y' = \frac{y}{x^2}.$$

Problem 3. Find the general solution of the differential equation:

$$y'' - 4y' + 3y = (-2x - 3)e^{3x}.$$

Problem 4. Find the general solution of the differential equation:

$$y' = \frac{y}{x} + \frac{x^2}{y^2}.$$

TOPIC No371224006
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Using the Laplace transform solve the Cauchy problem:

$$y'' + 2y' + y = e^{-2x}, \quad y(0) = y'(0) = 0.$$

Problem 2. Find the general solution of the differential equation:

$$y'' + y' - 2y = (-3x + 3) \cos 2x.$$

Problem 3. Find the general solution of the differential equation:

$$y' - 3y = \frac{-x + 2}{y^2} e^{9x}.$$

Problem 4. Find the general solution of the differential equation:

$$y' = \frac{x^3}{\sqrt{y}}.$$

TOPIC No371224007
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' + \frac{1}{x+1} y = -xy^2 .$$

Problem 2. Find the general solution of the differential equation:

$$y' = \frac{y^2}{1+x^2} .$$

Problem 3. Find the general solution of the differential equation:

$$y'' + 4y' + 3y = (-3x - 1) \cos 3x .$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y'' - 2y' + y = 9e^{-2x}, \quad y(0) = y'(0) = 0.$$

TOPIC No371224008
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y'' - 4y' + 4y = (3x - 2) \cos 2x .$$

Problem 2. Find the general solution of the differential equation:

$$y' + \frac{1}{2x+2} y = \frac{-x}{y^2} .$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y'' + y' - 6y = 25 e^{2x}, \quad y(0) = y'(0) = 0.$$

Problem 4. Find the general solution of the differential equation:

$$e^{2x} y' = y^2 .$$

TOPIC No371224009
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$yy' = 3.$$

Problem 2. Find the general solution of the differential equation:

$$y' = \frac{y}{x} - 3 \cotg \frac{y}{x}.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y''' + 2y'' + y' = 0, \quad y(0) = -1, \quad y'(0) = 2, \quad y''(0) = -1.$$

Problem 4. Find the general solution of the differential equation:

$$y'' - 3y' + 2y = (2x - 1) \sin x.$$

TOPIC No371224010
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' = \frac{y}{x} + 3 \cotg \frac{y}{x}.$$

Problem 2. Find the general solution of the differential equation:

$$y' = x^2 y.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y''' - 6y'' + 25y' = 0, \quad y(0) = -1, \quad y'(0) = -8, \quad y''(0) = -48.$$

Problem 4. Find the general solution of the differential equation:

$$y'' + 6y' + 9y = (-x + 1) \cos 3x.$$

TOPIC No371224011
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' - 3y = \frac{-x}{y^2} e^{9x}.$$

Problem 2. Find the general solution of the differential equation:

$$\sqrt{y} y' = 1.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y'' - 3y' = 9e^{3x}, \quad y(0) = y'(0) = 0.$$

Problem 4. Find the general solution of the differential equation:

$$y'' + y' - 2y = (-x - 1)e^{-2x}.$$

TOPIC No371224012
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' + \frac{1}{x-1} y = (2x+1)y^3.$$

Problem 2. Find the general solution of the differential equation:

$$y'' + 3y' + 2y = (-3x+1)e^{-2x}.$$

Problem 3. Find the general solution of the differential equation:

$$e^x y^2 y' = 3.$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y''' + y' = 0, \quad y(0) = 1, \quad y'(0) = 2, \quad y''(0) = 0.$$

TOPIC No371224013
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' + \frac{3}{2x-1} y = (-x+2)y^2.$$

Problem 2. Find the general solution of the differential equation:

$$y' - 3y^2 = 0.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y'' + 3y' + 2y = e^{-x}, \quad y(0) = y'(0) = 0.$$

Problem 4. Find the general solution of the differential equation:

$$y'' + 6y' + 9y = \frac{e^{-3x}}{x^3}.$$

TOPIC No371224015
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y'' - 5y' + 6y = 3xe^{2x}.$$

Problem 2. Find the general solution of the differential equation:

$$y' - y = -xe^{-2x}y^3.$$

Problem 3. Find the general solution of the differential equation:

$$\frac{y'}{\sqrt{x}} = y.$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y'' + 2y' + 5y = 10, \quad y(0) = y'(0) = 0.$$

TOPIC No371224016
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' = \frac{y}{x} - 3\sqrt[3]{\frac{x}{y}}.$$

Problem 2. Using the Laplace transform solve the Cauchy problem:

$$y'' - 9y = 36e^{-3x}, \quad y(0) = y'(0) = 0.$$

Problem 3. Find the general solution of the differential equation:

$$y' - y = 0.$$

Problem 4. Find the general solution of the differential equation:

$$y'' - y = (2x + 1)\cos x.$$

TOPIC No371224017
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' - \frac{1}{x+2}y = (x+1)\sqrt{y}.$$

Problem 2. Find the general solution of the differential equation:

$$y' = \sqrt{\frac{x}{y}}.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y'' - 9y = 36e^{-3x}, \quad y(0) = y'(0) = 0.$$

Problem 4. Find the general solution of the differential equation:

$$y'' + y' - 6y = (2x - 1)e^{-3x}.$$

TOPIC No371224018
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$\frac{y'}{e^{2x}} = 2\sqrt{y}.$$

Problem 2. Find the general solution of the differential equation:

$$y' + \frac{1}{2x} y = \frac{2x+1}{y}.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y''' - 2y'' + 10y' = 0, \quad y(0) = 1, \quad y'(0) = -2, \quad y''(0) = 16.$$

Problem 4. Find the general solution of the differential equation:

$$y'' - y = -3e^{-x}.$$

TOPIC No371224019
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Using the Laplace transform solve the Cauchy problem:

$$y'' - 8y' + 16y = e^{3x}, \quad y(0) = y'(0) = 0.$$

Problem 2. Find the general solution of the differential equation:

$$y'' + y' - 2y = (x - 2) \cos x .$$

Problem 3. Find the general solution of the differential equation:

$$3y' = \frac{\sin x}{y} .$$

Problem 4. Find the general solution of the differential equation:

$$y' - 3y = (x - 2) e^{-3x} y^2 .$$

TOPIC No371224020
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y'' + 4y = 3 \cos 2x .$$

Problem 2. Find the general solution of the differential equation:

$$2y' = x^3 y^2 .$$

Problem 3. Find the general solution of the differential equation:

$$y' + \frac{2}{x+1} y = \frac{2x+1}{y^2} .$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y''' - 2y'' = 0, \quad y(0) = 1, \quad y'(0) = 2, \quad y''(0) = -4.$$

TOPIC No371224021
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y'' + 5y' + 6y = (x - 2)e^{-3x}.$$

Problem 2. Using the Laplace transform solve the Cauchy problem:

$$y'' - 4y' + 13y = 9e^{2x}, \quad y(0) = y'(0) = 0.$$

Problem 3. Find the general solution of the differential equation:

$$\cos^2 x \, y' = y^2.$$

Problem 4. Find the general solution of the differential equation:

$$y' = \frac{y}{x} - \sqrt[3]{\frac{y}{x}}.$$

TOPIC No371224022
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y'' + y' - 6y = (2x - 3) \sin 2x .$$

Problem 2. Find the general solution of the differential equation:

$$y' = \frac{y}{x} - 3 \frac{x^2}{y^2} .$$

Problem 3. Find the general solution of the differential equation:

$$\frac{y'}{x} = 3y .$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y'' - 6y' + 9y = 25 e^{-2x}, \quad y(0) = y'(0) = 0.$$

TOPIC No371224023
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' = x\sqrt{y}.$$

Problem 2. Using the Laplace transform solve the Cauchy problem:

$$y''' - 4y'' + 4y' = 0, \quad y(0) = 1, \quad y'(0) = 2, \quad y''(0) = -4.$$

Problem 3. Find the general solution of the differential equation:

$$y' - 2y = (2x - 1)e^x\sqrt{y}.$$

Problem 4. Find the general solution of the differential equation:

$$2y''' + 3y'' = -3x^2 + x + 2.$$

TOPIC No371224024
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y'' + y' - 2y = 3e^x.$$

Problem 2. Find the general solution of the differential equation:

$$2y' = x^3 y^3.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y''' + 2y'' + y' = 0, \quad y(0) = 2, \quad y'(0) = 1, \quad y''(0) = -1.$$

Problem 4. Find the general solution of the differential equation:

$$y' + \frac{3}{x+2} y = x\sqrt{y}.$$

TOPIC No371224025
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y'' - 3y' + 2y = (3x + 3)e^x.$$

Problem 2. Find the general solution of the differential equation:

$$y' - y^2 = 0.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y''' + 2y'' + y' = 0, \quad y(0) = 1, \quad y'(0) = 3, \quad y''(0) = 1.$$

Problem 4. Find the general solution of the differential equation:

$$y' - \frac{1}{x-1} y = -2\sqrt{y}.$$

TOPIC No371224026
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' - \frac{1}{x+1} y = xy^3.$$

Problem 2. Using the Laplace transform solve the Cauchy problem:

$$y'' + 5y' + 6y = e^{-3x}, \quad y(0) = y'(0) = 0.$$

Problem 3. Find the general solution of the differential equation:

$$\sqrt{1-x^2} y' = \sqrt{y}.$$

Problem 4. Find the general solution of the differential equation:

$$y'' + 4y' + 3y = (-x - 2)e^{-x}.$$

TOPIC No371224027
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' = \frac{y}{x} + 3\sqrt[3]{\frac{x}{y}}.$$

Problem 2. Find the general solution of the differential equation:

$$\sqrt{1-x^2}y^2y' = 1.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y'' + 5y' + 6y = e^{-2x}, \quad y(0) = y'(0) = 0.$$

Problem 4. Find the general solution of the differential equation:

$$3y''' + 2y'' = 2x^2 - 2x - 3.$$

TOPIC No371224028
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' - \frac{1}{x}y = (-2x - 1)\sqrt{y}.$$

Problem 2. Using the Laplace transform solve the Cauchy problem:

$$y''' - 4y'' + 20y' = 0, \quad y(0) = 1, \quad y'(0) = 12, \quad y''(0) = 48.$$

Problem 3. Find the general solution of the differential equation:

$$y'' + 4y' + 3y = (-3x + 2)e^{-3x}.$$

Problem 4. Find the general solution of the differential equation:

$$3x^2y' = y^2.$$

TOPIC No371224029
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$\frac{y'}{\sin x} = 2\sqrt{y}.$$

Problem 2. Find the general solution of the differential equation:

$$y' = \frac{y}{x} + \operatorname{tg} \frac{y}{x}.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y''' + 4y' = 0, \quad y(0) = 3, \quad y'(0) = -6, \quad y''(0) = 0.$$

Problem 4. Solve the Cauchy problem:

$$y'' - 3y' + 2y = 2x + 3, \quad y(0) = -2, \quad y'(0) = 1.$$

TOPIC No371224030
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y'' - y = (-x + 2)e^x .$$

Problem 2. Using the Laplace transform solve the Cauchy problem:

$$y''' - 4y'' = 0, \quad y(0) = 0, \quad y'(0) = 3, \quad y''(0) = -16.$$

Problem 3. Find the general solution of the differential equation:

$$y' + \frac{3}{x-2} y = \frac{x+1}{y} .$$

Problem 4. Find the general solution of the differential equation:

$$y' - 2y = 0 .$$

TOPIC No371224031
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$\sqrt{1-x^2} y^2 y' = 3.$$

Problem 2. Find the general solution of the differential equation:

$$y' + \frac{3}{x-1} y = \frac{-2x+1}{y}.$$

Problem 3. Find the general solution of the differential equation:

$$y'' - y' - 2y = (x-3)e^{2x}.$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y''' + 4y'' = 0, \quad y(0) = -2, \quad y'(0) = 3, \quad y''(0) = 16.$$

TOPIC No371224032
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$3y' = \sqrt{x}y^2.$$

Problem 2. Find the general solution of the differential equation:

$$y'' - 3y' + 2y = e^x.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y'' + 16y = 16, \quad y(0) = y'(0) = 0.$$

Problem 4. Find the general solution of the differential equation:

$$y' + \frac{1}{x+2}y = \frac{-2x-2}{\sqrt{y}}.$$

TOPIC No371224033
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' + \frac{1}{x-1} y = \frac{-1}{\sqrt{y}}.$$

Problem 2. Find the general solution of the differential equation:

$$y'' - y = -2xe^{-x}.$$

Problem 3. Find the general solution of the differential equation:

$$\sqrt{1-x^2} y' = y^3.$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y''' - 6y'' + 10y' = 0, \quad y(0) = 2, \quad y'(0) = -1, \quad y''(0) = -6.$$

TOPIC No371224034
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y'' - 2y' - 3y = (2x - 1)e^{3x}.$$

Problem 2. Find the general solution of the differential equation:

$$y' - \frac{1}{x+2}y = \frac{-2x-2}{y^2}.$$

Problem 3. Find the general solution of the differential equation:

$$\frac{y'}{e^{3x}} = 3y^2.$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y''' + 6y'' + 9y' = 0, \quad y(0) = 1, \quad y'(0) = 0, \quad y''(0) = -9.$$

TOPIC No371224035
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y'' + 4y' + 4y = (-2x + 3)e^{-2x}.$$

Problem 2. Using the Laplace transform solve the Cauchy problem:

$$y''' + 8y'' + 16y' = 0, \quad y(0) = 2, \quad y'(0) = 8, \quad y''(0) = -16.$$

Problem 3. Find the general solution of the differential equation:

$$y' - y = \frac{-2x + 2}{y} e^{2x}.$$

Problem 4. Find the general solution of the differential equation:

$$y' = \frac{x^3}{y}.$$

TOPIC No371224036
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' + \frac{2}{x+2} y = \frac{2x-1}{\sqrt{y}} .$$

Problem 2. Using the Laplace transform solve the Cauchy problem:

$$y''' + 3y'' = 0, \quad y(0) = 0, \quad y'(0) = 1, \quad y''(0) = -9.$$

Problem 3. Find the general solution of the differential equation:

$$\sqrt{1-x^2} y y' = 1 .$$

Problem 4. Find the general solution of the differential equation:

$$y'' + 2y' + y = \frac{e^{-x}}{x} .$$

TOPIC No371224037
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' + \frac{2}{x-2}y = (2x+1)y^3.$$

Problem 2. Using the Laplace transform solve the Cauchy problem:

$$y'' - 2y' + y = e^{2x}, \quad y(0) = y'(0) = 0.$$

Problem 3. Find the general solution of the differential equation:

$$yy' = 2.$$

Problem 4. Find the general solution of the differential equation:

$$y'' + 4y = \frac{2}{\sin 2x}.$$

TOPIC No371224038
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Using the Laplace transform solve the Cauchy problem:

$$y'' - 2y' + 17y = 68, \quad y(0) = y'(0) = 0.$$

Problem 2. Find the general solution of the differential equation:

$$y'' + 4y' + 3y = (-3x + 1) \sin 3x.$$

Problem 3. Find the general solution of the differential equation:

$$3y' = xy.$$

Problem 4. Find the general solution of the differential equation:

$$y' = \frac{y}{x} - \frac{y^3}{x^3}.$$

TOPIC No371224039
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' = \frac{y}{x} + 2\sqrt{\frac{x}{y}}.$$

Problem 2. Find the general solution of the differential equation:

$$y'' - 3y' + 2y = (-x - 1)\sin 2x.$$

Problem 3. Find the general solution of the differential equation:

$$y' - y^2 = 0.$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y''' - 4y'' + 4y' = 0, \quad y(0) = 2, \quad y'(0) = 4, \quad y''(0) = 4.$$

TOPIC No373224040
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' + \frac{1}{x+1} y = 2\sqrt{y}.$$

Problem 2. Find the general solution of the differential equation:

$$y''' + y'' = 2x^2 + 3x + 2.$$

Problem 3. Find the general solution of the differential equation:

$$\sqrt{1-x^2} y' = y^3.$$

Problem 4. Using the Laplace transform solve the Cauchy problem:

$$y''' - 2y'' = 0, \quad y(0) = 2, \quad y'(0) = -3, \quad y''(0) = 4.$$

TOPIC No373224041
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$\frac{y'}{x} = 2y.$$

Problem 2. Find the general solution of the differential equation:

$$y' + \frac{3}{x+2}y = \frac{-2x+2}{y^2}.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y'' + 6y' + 25y = 32e^x, \quad y(0) = y'(0) = 0.$$

Problem 4. Find the general solution of the differential equation:

$$y'' - 3y' + 2y = (-x - 2)e^{2x}.$$

TOPIC No373224042
MATHEMATICAL MODELLING AND DIFFERENTIAL EQUATIONS

Problem 1. Find the general solution of the differential equation:

$$y' = \frac{y}{x} + \operatorname{tg} \frac{y}{x}.$$

Problem 2. Find the general solution of the differential equation:

$$y^3 y' = \cos x.$$

Problem 3. Using the Laplace transform solve the Cauchy problem:

$$y''' - 2y'' + y' = 0, \quad y(0) = 0, \quad y'(0) = -2, \quad y''(0) = -1.$$

Problem 4. Find the general solution of the differential equation:

$$y'' + y' - 2y = (-x - 1)e^x.$$